

Alex Kaiser

STAFF EMBEDDED SOFTWARE ENGINEER

Raleigh, NC
www.alexjkaiser.com

630-487-0872 alexkaiser@me.com
in/alex-kaiser34 alexkaiser34

PROFESSIONAL SUMMARY

Staff Embedded Software Engineer with 4+ years of experience spanning hardware, FPGA, and embedded systems, along with full-stack application development and a growing focus on AI. Skilled at translating complex requirements into efficient, reliable software and supporting the full development lifecycle.

EXPERIENCE

Staff Embedded Software Engineer Dec 2025 – Present
Itron Inc. Raleigh, NC

- Lead embedded software, internal tooling, and AI applications for smart electric metering.
- Developed AI applications with RAG-based pipelines and internally hosted LLMs (LangChain, Ollama).
 - Implemented performance optimizations delivering >10% CPU reduction, lowering cost and operational risk.
 - Revamped internal tools to accelerate team workflows and reduce turnaround time.
 - Continued embedded software development for electric meters on the core C++/SQLite stack.

Embedded Software Engineer Apr 2024 – Dec 2025
Itron Inc. Raleigh, NC

- Embedded software engineering for smart electric metering.
- Optimized embedded software with C++ and SQLite while ensuring performance and design compliance.
 - Resolved critical software issues through root-cause analysis, improving system stability.
 - Enhanced CPU efficiency by profiling and refactoring code, presenting optimizations to leadership.
 - Developed and optimized desktop applications in ASP.NET, improving functionality and performance.

Software Engineer Co-op, FPGA / Secure Tech May 2021 – Aug 2023
DornerWorks Ltd. Grand Rapids, MI

- FPGA and secure-technology engineering across embedded and full-stack systems.
- Managed a team of engineers to design an application demonstrating hypervisor benefits.
 - Designed embedded hardware and software drivers using VHDL, C, Python, and the AMBA AXI protocol.
 - Accelerated machine learning algorithms by 100x through collaborative optimization.
 - Built full-stack desktop applications with React, TypeScript, and Python, enhancing UX.

EDUCATION

B.S. Electrical Engineering · B.S. Computer Engineering Aug 2018 – Jan 2024
Grand Valley State University Allendale, MI

- 3.9 GPA · Dean's List every semester
- Frederik Meijer Honors College · Tau Beta Pi Engineering Honors Society
- Minor in Business Administration

PROJECTS

Embedded CPU Code Profiler Jun 2025 – Oct 2025

- Designed and implemented a function-level profiling approach for memory constrained Itron embedded devices.
- Built and validated the profiler to generate call-graph metrics (call counts, total time, average time).
- Created a companion visualization app so developers could quickly identify CPU hotspots.

ASP.NET Core Itron ToolSuite Apr 2024 – Present

- Own the internal ASP.NET Core ToolSuite, used by thousands of developers to interact with products and data.
- Lead feature development and refactoring across the suite to expand functionality and improve maintainability.
- Coordinate cross-functional requirements and established a CI pipeline to automate builds and deployments.

SKILLS

Languages

C/C++C#PythonJavaScript/TypeScriptVHDL/VerilogSQL/NoSQLBash/Shell

Frameworks & AI

ReactASP.NET CoreLangChainOllamaRAG PipelinesSelf-Hosted LLMs

Platforms & Tools

DockerAWSGitCI/CDSQLiteQdrantPostgres

Professional

Embedded System DesignHardware/FPGA DesignFull-Stack DevelopmentAI System DesignTeam LeadershipProject ManagementCustomer Interaction